

Derrick Chen

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EDUCATION

Georgia Institute of Technology

Expected May 2028

Bachelor of Science in Computer Science

- Relevant Coursework: Data Structures and Algorithms, Artificial Intelligence, Object Oriented Programming, Computer Architecture, Machine Learning, Operating Systems, Database Design

EXPERIENCE

Software Engineer Co-Op

Aug 2026 - Present

IBM

San Jose, CA

- ARIA (Autonomous Reasoning Intelligent Agents) - Framework for building conversation agents for IBM Cloud

Software Engineer Intern

June 2026 - Aug 2026

Walmart

Bentonville, AR

- Designed an event-driven stocking confirmation flow processing **3,000+ requests/minute**, replacing inventory guesswork with real-time stocking updates saving a store hundreds of thousands annually
- Implemented the backend with Java, Spring Boot, and Apache Kafka, implementing event mapping, validation, and publishing logic with **90%+ code coverage**
- Built a Grafana and Prometheus/Apache JMeter dashboard tracking **p95/p99 latency**, HTTP throughput, Kafka publish rates, and CPU/heap utilization to monitor production metrics

Software Engineer Intern

June 2025 - Aug 2025

BigID

Atlanta, GA

- Developed integration for **BigID customers** to stream DSPM cases into **AWS Security Lake** using **IAM** and **S3**, allowing customers to centralize DSPM data within their AWS environments
- Built the backend in **Node.js**, using ETL pipelines to handle terabytes of security data, cutting ingestion time by over **100x** (hours to minutes)
- Wrote **Jest unit tests** achieving **80%+ code coverage** and built end-to-end integration tests for DSPM - AWS workflows; used **Jenkins** to manage test environments and automate cloud deployments

Undergraduate Researcher

Aug 2024 - Dec 2024

GROWER Research Lab

Atlanta, GA

- Wrote **Python** scripts to web scrape data on power outages from 13+ sources, expanding my team's dataset coverage by **15%** of U.S. states
- Cleaned and analyzed datasets with **1M+ records**, enabling data analysis in emergency response, public investments, and public policy
- Collaborated on a subteam of 6 using Agile methodologies such as daily standups, sprint planning, and pair programming, while partnering with Oak Ridge National Laboratory

PROJECTS

Referee Ruling Engine | *Python, FastAPI, Next.js, TypeScript, PostgreSQL, pgvector*

ask-the-ref.com

- Built an NBA rules verification engine averaging **1.6K users/visits monthly** that translates **natural language** basketball scenarios into a legal or illegal verdict
- Designed a hybrid **RAG retrieval system** combining **pgvector similarity search** and **PostgreSQL full-text search** to surface the most relevant rule sections for each scenario
- Engineered a structured reasoning pipeline that extracts scenario facts such as **timing, possession, location, and intent**, then evaluates them exclusively against retrieved official rules to reduce unsupported model judgments

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, TypeScript, C, C++

Frameworks: React, Node.js, Next.js, Flask, Django, Spring Boot, Express

Developer Tools: AWS, Git/GitHub, Vercel, Jira, Playwright, Azure, PyTest, Docker, Kafka

Databases: PostgreSQL, MongoDB, MySQL

AI Development: Model Context Protocol (MCP), AI Agents, Claude Code, Cursor, Prompt Engineering

EXTRA

Awards: Coca-Cola First Generation Scholar, Zell Miller Scholarship Recipient

Extracurriculars: WebDev @ GT, Intramural Sports, Blockchain @ GT, Greek Life (Director of Career Development)